Summary Sheet

ProteinVision

MACCS Keys

Quick reference for constructing and understanding MACCS keys fingerprints.

MACCS keys encode a molecule by answering a fixed checklist of structural questions (see Appendix). The commonly used 166-key version has one position per question in a binary vector, a sequence of 0s and 1s. The same questions occupy the same positions for every molecule.

1. Start from the molecular graph

- Represent atoms as nodes and chemical bonds as edges. Keep track of attached hydrogens.

2. Initialize the fingerprint

- Start with 166 zeros. Each position is assigned to a predefined key.

3. Evaluate the structural questions

- Check every key against the molecular graph using that key's definition. Keys can test atom types, bond types, ring features, connected patterns, or count thresholds.
- A pattern only needs to occur somewhere in the molecule; it need not describe the entire structure.

4. Record the binary answers

- Set the corresponding bit to 1 if the condition is satisfied; otherwise leave it at 0.
- **Repeated matches:** Finding the same pattern several times does not increase its bit beyond 1.
- **Count thresholds:** Some keys ask whether a count meets a specified threshold. They preserve limited count information, but each answer remains binary.
- **Overlapping features:** The same atoms and bonds can satisfy several keys.

5. Read the fingerprint and understand its limits

- Each position has a known meaning, so a set bit can be interpreted directly from its key definition.
- The fingerprint gives a compact description of selected structural features. It does not preserve the complete molecular structure, and generally does not retain exact feature counts.
- Different molecules can have identical fingerprints if all key answers agree. Identical fingerprints therefore do not establish that two molecules are identical.

Appendix

1. Is the reserved isotope-related key active?
2. Is there a rutherfordium atom (atomic number 104)?
3. Is there a germanium, arsenic, selenium, tin, antimony, tellurium, lead, bismuth, or polonium atom?
4. Is there an actinide atom, from actinium through lawrencium?
5. Is there a scandium, titanium, yttrium, zirconium, or hafnium atom?
6. Is there a lanthanide atom, from lanthanum through lutetium?
7. Is there a vanadium, chromium, manganese, niobium, molybdenum, technetium, tantalum, tungsten, or rhenium atom?
8. Is there a four-membered ring containing at least one atom other than carbon or hydrogen?
9. Is there an iron, cobalt, nickel, ruthenium, rhodium, palladium, osmium, iridium, or platinum atom?
10. Is there a beryllium, magnesium, calcium, strontium, barium, or radium atom?
11. Is there a four-membered ring?
12. Is there a copper, zinc, silver, cadmium, gold, or mercury atom?
13. Is there a nitrogen atom bonded to one oxygen atom and two carbon atoms?
14. Is there a sulfur–sulfur single bond?
15. Is there a carbon atom bonded to three oxygen atoms?
16. Is there a three-membered ring containing at least one atom other than carbon or hydrogen?
17. Is there a carbon–carbon triple bond?
18. Is there a boron, aluminium, gallium, indium, or thallium atom?
19. Is there a seven-membered ring?
20. Is there a silicon atom?
21. Is there a carbon–carbon double bond with one of its carbon atoms also bonded to two atoms other than carbon or hydrogen?
22. Is there a three-membered ring?
23. Is there a carbon atom bonded to one nitrogen atom and two oxygen atoms?
24. Is there a nitrogen–oxygen single bond?
25. Is there a carbon atom bonded to three nitrogen atoms?
26. Is there a carbon–carbon double bond in a ring, with one of the double-bonded carbons also involved in two additional ring bonds?
27. Is there an iodine atom?
28. Is there a three-atom path in which a CH_2 carbon is between two atoms other than carbon or hydrogen?
29. Is there a phosphorus atom?
30. Is there an atom other than carbon or hydrogen bonded to three carbon atoms and a fourth atom?

31. Is there a bond between an atom other than carbon or hydrogen and a fluorine, chlorine, bromine, or iodine atom?

32. Is there a three-atom path consisting of carbon, sulfur, and nitrogen, in that order?

33. Is there a nitrogen–sulfur bond of any type?

34. Is there a nonaromatic CH_2 carbon double-bonded to another atom?

35. Is there a lithium, sodium, potassium, rubidium, caesium, or francium atom?

36. Is there a sulfur atom in a ring?

37. Is there a carbon atom bonded to two nitrogen atoms and one oxygen atom?

38. Is there a carbon atom bonded to two nitrogen atoms and another carbon atom?

39. Is there a sulfur atom bonded to three oxygen atoms?

40. Is there a sulfur–oxygen single bond?

41. Is there a carbon–nitrogen triple bond?

42. Is there a fluorine atom?

43. Is there a three-atom path in which both end atoms are neither carbon nor hydrogen and both have at least one attached hydrogen?

44. Is there an atom other than hydrogen, carbon, nitrogen, oxygen, fluorine, silicon, phosphorus, sulfur, chlorine, bromine, or iodine?

45. Is there a carbon–carbon double bond with a nitrogen atom also bonded to one of the double-bonded carbons?

46. Is there a bromine atom?

47. Is there a three-atom path consisting of sulfur, any atom, and nitrogen?

48. Is there an atom other than carbon or hydrogen bonded to three oxygen atoms?

49. Does any atom carry a nonzero formal charge?

50. Is there a carbon–carbon double bond with one of its carbon atoms also bonded to two further carbon atoms?

51. Is there a three-atom path consisting of carbon, sulfur, and oxygen?

52. Is there a nitrogen–nitrogen bond of any type?

53. Is there a five-atom path whose two end atoms are neither carbon nor hydrogen and both have at least one attached hydrogen?

54. Is there a four-atom path whose two end atoms are neither carbon nor hydrogen and both have at least one attached hydrogen?

55. Is there a three-atom path consisting of oxygen, sulfur, and oxygen?

56. Is there a nitrogen atom bonded to two oxygen atoms and one carbon atom?

57. Is there an oxygen atom in a ring?

58. Is there a three-atom path with sulfur between two atoms other than carbon or hydrogen?

59. Is a sulfur atom bonded by a nonaromatic bond to an atom that also has an aromatic bond to another atom?

60. Is there a sulfur–oxygen double bond?

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61. Does a sulfur atom have at least three neighboring atoms?

62. Is there a four-atom path whose first and last bonds belong to rings, but whose middle bond does not?

63. Is there a nitrogen–oxygen double bond?

64. Is a sulfur atom connected by a non-ring bond to an atom that is also part of a ring bond?

65. Is there an aromatic bond between an aromatic carbon and an aromatic nitrogen?

66. Is there a carbon atom bonded to three carbon atoms and a fourth atom?

67. Is there a bond between sulfur and an atom other than carbon or hydrogen?

68. Are there two directly bonded atoms, neither carbon nor hydrogen, that each have at least one attached hydrogen?

69. Are there two directly bonded atoms, neither carbon nor hydrogen, with at least one of them carrying an attached hydrogen?

70. Is there a three-atom path with nitrogen between two atoms other than carbon or hydrogen?

71. Is there a nitrogen–oxygen bond of any type?

72. Is there a four-atom path with oxygen atoms at both ends?

73. Is there a sulfur atom double-bonded to any atom?

74. Are two nonaromatic CH_3 carbon atoms bonded to the same intervening atom?

75. Is there a nitrogen atom joined to one neighboring atom by a non-ring bond and another by a ring bond?

76. Is there a carbon–carbon double bond with one of its carbon atoms also bonded to two additional atoms?

77. Is there a three-atom path with nitrogen atoms at both ends?

78. Is there a carbon–nitrogen double bond?

79. Is there a four-atom path with nitrogen atoms at both ends?

80. Is there a five-atom path with nitrogen atoms at both ends?

81. Is there a sulfur atom bonded to another atom that has at least two additional neighbors?

82. Is there a three-atom path in which a CH_2 carbon connects any atom to an atom other than carbon or hydrogen that carries at least one hydrogen?

83. Is there a five-membered ring containing at least one atom other than carbon or hydrogen?

84. Is there a nonaromatic nitrogen atom with exactly two attached hydrogens?

85. Is there a nitrogen atom bonded to at least three carbon atoms?

86. Is there an atom other than carbon or hydrogen connected to two nonaromatic carbons that each bear two or three hydrogens, with both connections being single or aromatic bonds?

87. Is a fluorine, chlorine, bromine, or iodine atom attached through a non-ring bond to an atom involved in a ring bond?

88. Is there a sulfur atom?

89. Is there a five-atom path with oxygen atoms at both ends?

90. Is there a five-atom path beginning with a hydrogen-bearing atom other than carbon or hydrogen and having CH_2 in position four; a four-membered ring in which such a hydrogen-bearing atom is next to CH_2; or a three-membered ring containing CH_2 with such a hydrogen-bearing atom attached to another ring atom?

91. Is there a six-atom path beginning with a hydrogen-bearing atom other than carbon or hydrogen and having CH_2 in position five; a five-membered ring in which such a hydrogen-bearing atom is next to CH_2; a four-membered ring with CH_2 next to a ring atom to which such a hydrogen-bearing atom is directly attached; or a three-membered ring containing CH_2 with such a hydrogen-bearing atom connected to another ring atom through one intervening atom?

92. Is there a carbon atom bonded to an oxygen atom, a nitrogen atom, and another carbon atom?

93. Is there a bond between a nonaromatic CH_3 carbon and an atom other than carbon or hydrogen?

94. Is there a bond between nitrogen and an atom other than carbon or hydrogen?

95. Is there a four-atom path with nitrogen at one end and oxygen at the other?

96. Is there a five-membered ring?

97. Is there a five-atom path with nitrogen at one end and oxygen at the other?

98. Is there a six-membered ring containing at least one atom other than carbon or hydrogen?

99. Is there a carbon–carbon double bond?

100. Is there a three-atom path in which a CH_2 carbon connects any atom to nitrogen?

101. Is there a ring containing between 8 and 14 atoms, inclusive?

102. Is there a bond between oxygen and an atom other than carbon or hydrogen?

103. Is there a chlorine atom?

104. Is there a four-atom path that starts with a hydrogen-bearing atom other than carbon or hydrogen and has a CH_2 carbon in the third position?

105. Is there an atom participating in at least three ring bonds?

106. Is there an atom bonded to three atoms other than carbon or hydrogen?

107. Is there a fluorine, chlorine, bromine, or iodine atom bonded to another atom that has at least two additional neighbors?

108. Is there a six-atom path starting at a CH_3 carbon and containing a CH_2 carbon in its fifth position?

109. Is there a three-atom path consisting of any atom, a CH_2 carbon, and oxygen?

110. Is there a three-atom path consisting of nitrogen, carbon, and oxygen?

111. Is there a four-atom path starting at nitrogen and containing a CH_2 carbon in its third position?

112. Does any atom have at least four neighboring atoms?

113. Is an oxygen atom bonded by a nonaromatic bond to an atom that also has an aromatic bond to another atom?

114. Is there a three-atom path beginning with a CH_3 carbon followed by a CH_2 carbon?

115. Is there a four-atom path beginning with a CH_3 carbon and containing a CH_2 carbon in its third position?

116. Is there a five-atom path starting at a CH_3 carbon and having a CH_2 carbon in position four, or a CH_3 carbon attached to a three-membered ring containing a CH_2 carbon?

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117. Is there a three-atom path with nitrogen at one end and oxygen at the other?

118. Are there at least two distinct atoms, each of which either begins a four-atom path with two CH_2 carbons in the middle or belongs to a three-membered ring whose other two atoms are CH_2 carbons?

119. Is there a nitrogen atom double-bonded to any atom?

120. Are at least two ring atoms elements other than carbon?

121. Is there a nitrogen atom in a ring?

122. Does a nitrogen atom have at least three neighboring atoms?

123. Is there a three-atom path consisting of oxygen, carbon, and oxygen?

124. Are two atoms other than carbon or hydrogen bonded to one another?

125. Are there at least two rings whose constituent bonds are all aromatic?

126. Is there an oxygen atom connected to two neighbors by non-ring bonds?

127. Are there at least two occurrences of a ring bond followed by a non-ring bond to an oxygen atom?

128. Is there a seven-atom path with CH_2 carbons in positions two and six; a six-membered ring with two CH_2 carbons next to the same ring atom; a four-membered ring with a CH_2 ring atom next to a ring atom attached to an outside CH_2 that has another neighbor; or a three-membered ring with a CH_2 ring atom next to a ring atom connected through one other atom to an outside CH_2 that has another neighbor?

129. Is there a six-atom path with CH_2 carbons in positions two and five; a five-membered ring with two CH_2 carbons next to the same ring atom; or a three-membered ring with a CH_2 ring atom next to a ring atom attached to an outside CH_2 that has another neighbor?

130. Are there at least two distinct bonds between atoms other than carbon or hydrogen?

131. Are there at least two atoms other than carbon or hydrogen that each have at least one attached hydrogen?

132. Is there a four-atom path starting at oxygen and containing a CH_2 carbon in its third position?

133. Is there a ring bond followed by a non-ring bond to nitrogen?

134. Is there a fluorine, chlorine, bromine, or iodine atom?

135. Is a nitrogen atom bonded by a nonaromatic bond to an atom that also has an aromatic bond to another atom?

136. Are there at least two matches of an oxygen atom double-bonded to another atom?

137. Is there an atom other than carbon in a ring?

138. Are there at least two matches of a three-atom path starting with an atom other than carbon or hydrogen and continuing through a CH_2 carbon?

139. Is there a nonaromatic oxygen atom with at least one attached hydrogen?

140. Are there at least four oxygen atoms?

141. Are there at least three nonaromatic carbon atoms each bearing exactly three hydrogens?

142. Are there at least two nitrogen atoms?

143. Is there a ring bond followed by a non-ring bond to oxygen?

144. Is there a four-atom path whose successive bonds are nonaromatic, aromatic, and nonaromatic?

145. Are there at least two distinct sets of six atoms that each form a ring?

146. Are there at least three oxygen atoms?

147. Is there a four-atom path whose two middle atoms are CH_2 carbons, or a three-membered ring containing two CH_2 carbons?

148. Does an atom other than carbon or hydrogen have at least three neighboring atoms?

149. Are there at least two nonaromatic carbons each bearing three or four hydrogens?

150. Is there a four-atom path whose successive bonds are non-ring, ring, and non-ring?

151. Is there a nitrogen atom (aromatic or nonaromatic) with at least one attached hydrogen?

152. Is there a carbon atom bonded to oxygen and two other carbon atoms?

153. Is there a three-atom path starting with an atom other than carbon or hydrogen and continuing through a CH_2 carbon?

154. Is there a carbon–oxygen double bond?

155. Is there a CH_2 carbon connected to two atoms through bonds that are not part of rings?

156. Is there a nitrogen atom bonded to another atom that has at least two additional neighbors?

157. Is there a carbon–oxygen single bond?

158. Is there a carbon–nitrogen single bond?

159. Are there at least two oxygen atoms?

160. Is there a nonaromatic carbon atom bearing three or four hydrogens?

161. Is there a nitrogen atom?

162. Is there an aromatic atom?

163. Is there a six-membered ring?

164. Is there an oxygen atom?

165. Is there an atom that belongs to a ring?

166. Does the molecule consist of at least two disconnected components?

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